

Data Science - Assignment 2

Submission Requirement

- Submit **one** `.ipynb` **notebook**
- Use:
 - Code cells for implementation
 - Markdown cells for explanations and answers
- Your notebook should be well-organized and readable

Background

You are working as a data scientist for a wine company.

Your task is to analyze the **Wine Quality Dataset** to understand:

- How different chemical properties behave
- Whether there are unusual observations (outliers)
- How the data is distributed

Your analysis will help the company better understand wine characteristics and quality patterns.

You can download the dataset using this link (<https://github.com/XiaochenLi-w/CSC-405-605-705-Data-Science-Fall-2026/blob/main/dataset/WineQT.csv>) or from Kaggle (<https://www.kaggle.com/datasets/yasserh/wine-quality-dataset>).

Part 1: Data Exploration (Descriptive Statistics) — 50 points

Task 1: Load the Dataset

- Load the **Wine Quality Dataset** using `pandas`

Task 2: Statistical Summary

Select **three variables of your choice** from the dataset.

For each variable, compute:

- mean
- median
- variance
- standard deviation
- minimum
- maximum

Write Code and Answer Questions

1. Which variable has the largest variance? What does this imply about the data?
2. Which variable appears the most stable? Why?

Task 3: Z-score & Outlier Detection

Select **one variable** (from the three you chose).

1. Compute the z-score for every data point:

$$z = \frac{x - \mu}{\sigma}$$

1. Apply the 3σ rule to identify potential outliers.

Write Code and Answer Questions

1. How many potential outliers did you detect? Please include some explanations.
2. Do you think these outliers should be removed in a data science pipeline? Briefly explain.

Part 2: Kernel Density Estimation (KDE) — 50 points

You will now analyze how a variable is distributed using Kernel Density Estimation (KDE).

Task 1: KDE Modeling

- Select **one continuous variable** from Wine Quality Dataset.
- Use a Python library (e.g., `sklearn.neighbors.KernelDensity`) to:
 - Fit a KDE model
 - Choose a kernel function (e.g., Gaussian, Epanechnikov, uniform)
 - Choose a bandwidth

Task 2: Evaluate Density

Compute the estimated density at the following points:

$$x = [\mu - 2\sigma, \mu - \sigma, \mu, \mu + \sigma, \mu + 2\sigma]$$

Report your results in a table:

x value	Estimated density
...	...

Write Code and Answer Questions

1. What kernel function did you choose?
2. How does the density change as x moves away from the mean?
3. Based on your results, where is the data most concentrated?

Bonus (Optional, +10 points)

- Try a different bandwidth
- Compare the density values

Question:

- What effect does bandwidth have on the results?

Academic Integrity

Each student is expected to complete this assignment independently.

- You may choose different variables, which is encouraged.
- If multiple students select the same variables for all tasks, submissions may be compared line-by-line.